1. What does CSS stand for in web development?

(A) Cascading Style Sheets
(B) Computer Style Syntax
(C) Creative Styling System
(D) Colorful Standard Sheets

2. Which method of applying CSS is embedded directly inside an HTML tag using the style attribute?

(A) Inline CSS
(B) Internal CSS
(C) External CSS
(D) Imported CSS

3. Internal CSS styling rules are written inside which HTML tag within the <head> section?

(A) <style>
(B) <css>
(C) <script>
(D) <link>

4. Which HTML tag is used to reference an external CSS stylesheet file?

(A) <link rel="stylesheet" href="styles.css">
(B) <style src="styles.css">
(C) <css href="styles.css">
(D) <include file="styles.css">

5. In CSS syntax, a rule consists of a selector and a declaration block enclosed in:

(A) Curly braces { }
(B) Square brackets [ ]
(C) Parentheses ( )
(D) Angle brackets < >

6. Which character is used in CSS selectors to target an element by its ID?

(A) # (hash symbol)
(B) . (dot symbol)
(C) * (asterisk)
(D) @ (at symbol)

7. Which character is used in CSS selectors to target elements by their class name?

(A) . (dot symbol)
(B) # (hash symbol)
(C) & (ampersand)
(D) $ (dollar sign)

8. The universal selector in CSS that matches every element on the page is:

(A) * (asterisk)
(B) all
(C) %
(D) #

9. Which CSS property is used to change the background color of an element?

(A) background-color
(B) color
(C) bgcolor
(D) surface-color

10. Which CSS property is used to change the text color of an element?

(A) color
(B) text-color
(C) font-color
(D) foreground

11. Which CSS property sets the typeface or font family of text?

(A) font-family
(B) typeface
(C) font-name
(D) text-style

12. Which CSS property controls the size of text typography?

(A) font-size
(B) text-size
(C) font-scale
(D) type-size

13. Which CSS property is used to make text bold?

(A) font-weight: bold;
(B) font-style: bold;
(C) text-bold: true;
(D) text-weight: 700;

14. Which CSS property makes text italicized?

(A) font-style: italic;
(B) font-weight: italic;
(C) text-slant: italic;
(D) font-variant: italic;

15. Which CSS property controls the horizontal alignment of text within a block?

(A) text-align
(B) horizontal-align
(C) align-content
(D) text-justify

16. Which CSS property removes underlines from hyperlinks?

(A) text-decoration: none;
(B) text-underline: none;
(C) link-style: plain;
(D) font-decoration: off;

17. The four concentric rectangular layers comprising the CSS Box Model, from innermost to outermost, are:

(A) Content, Padding, Border, Margin
(B) Content, Border, Padding, Margin
(C) Margin, Border, Padding, Content
(D) Content, Margin, Border, Padding

18. Which box model layer clears space around the content, inside of any defined border?

(A) Padding
(B) Margin
(C) Border
(D) Outline

19. Which box model layer provides transparent spacing outside of the border between adjacent elements?

(A) Margin
(B) Padding
(C) Border
(D) Content

20. In standard CSS shorthand (e.g., 'padding: 10px 20px 30px 40px;'), what is the clockwise sequence of sides?

(A) Top, Right, Bottom, Left (TRBL)
(B) Top, Left, Bottom, Right
(C) Left, Top, Right, Bottom
(D) Bottom, Right, Top, Left

21. Which CSS property controls whether an element is treated as block, inline, flex, grid, or hidden?

(A) display
(B) position
(C) visibility
(D) float

22. What is the default display value of a element in HTML?

(A) block
(B) inline
(C) inline-block
(D) flex

23. What is the default display value of a element in HTML?

(A) inline
(B) block
(C) contents
(D) table-cell

24. An element with 'display: none;' differs from 'visibility: hidden;' because 'display: none;':

(A) Removes the element entirely from the document layout flow, taking up zero visual space
(B) Leaves an empty space where the element was
(C) Makes it semi-transparent
(D) Disables JavaScript click events only

25. Which CSS property controls the layering order of overlapping positioned elements along the z-axis?

(A) z-index
(B) stack-order
(C) elevation
(D) layer-index

26. The 'z-index' property only functions on elements whose CSS 'position' property is set to something other than:

(A) static
(B) relative
(C) absolute
(D) fixed

27. An element with 'position: relative;' is positioned relative to:

(A) Its normal position in the default document layout flow
(B) The browser viewport
(C) The nearest ancestor with position: absolute
(D) The root <html> tag

28. An element with 'position: absolute;' is positioned relative to:

(A) Its nearest ancestor that has a position other than static (or the initial containing block)
(B) Its parent always, regardless of parent position
(C) The browser window viewport always
(D) The screen display coordinates

29. An element with 'position: fixed;' is positioned relative to:

(A) The browser viewport, remaining anchored in place even when the page is scrolled
(B) Its parent container
(C) The entire HTML document body
(D) The mouse cursor

30. Which CSS display mode activates the Flexible Box (Flexbox) 1-dimensional layout model?

(A) display: flex;
(B) display: box;
(C) display: grid;
(D) display: flexible;

31. Which Flexbox property defines whether items are laid out horizontally in a row or vertically in a column?

(A) flex-direction
(B) flex-wrap
(C) justify-content
(D) align-items

32. Which Flexbox property aligns flex items along the primary Main Axis?

(A) justify-content
(B) align-items
(C) align-content
(D) flex-align

33. Which Flexbox property aligns flex items along the perpendicular Cross Axis?

(A) align-items
(B) justify-content
(C) flex-direction
(D) align-self

34. Which Flexbox property determines whether flex items are forced onto a single line or can wrap onto multiple lines?

(A) flex-wrap
(B) flex-flow
(C) flex-shrink
(D) flex-basis

35. Which CSS property activates the 2-dimensional Grid Layout module?

(A) display: grid;
(B) display: flex;
(C) display: table;
(D) display: columns;

36. In CSS Grid, which property defines the number and sizing widths of vertical columns?

(A) grid-template-columns
(B) grid-column-gap
(C) grid-template-rows
(D) grid-auto-flow

37. Which flexible CSS unit introduced with CSS Grid represents a fraction of the available free space in the grid container?

(A) fr (fractional unit)
(B) em
(C) rem
(D) vh

38. Which property sets the gutter gap spacing between grid rows and columns?

(A) gap (or row-gap / column-gap)
(B) grid-margin
(C) gutter
(D) cell-spacing

39. Which CSS feature allows styling rules to apply selectively based on device characteristics like screen width or orientation?

(A) Media Queries (@media)
(B) CSS Variables
(C) CSS Transforms
(D) CSS Animations

40. Which pseudo-class in CSS applies styles when a user hovers their mouse pointer over an element?

(A) :hover
(B) :active
(C) :focus
(D) :visited

41. What is the critical difference between 'box-sizing: content-box' (default) and 'box-sizing: border-box'?

(A) border-box includes padding and border within the declared width and height, preventing layout breakage
(B) content-box includes margins
(C) border-box disables padding
(D) border-box only works on tables

42. If a box has width: 200px, padding: 20px, and border: 5px, what is its total rendered horizontal width under 'box-sizing: content-box'?

(A) 250px (200 + 20*2 + 5*2)
(B) 200px
(C) 225px
(D) 240px

43. For the box in Q42, what is its total rendered horizontal width under 'box-sizing: border-box'?

(A) 200px (padding and border are absorbed inside the 200px)
(B) 250px
(C) 225px
(D) 210px

44. What is CSS Specificity hierarchy from highest precedence to lowest among normal declarations?

(A) !important > Inline styles > IDs > Classes/Attributes/Pseudo-classes > Elements/Pseudo-elements
(B) IDs > Classes > Inline > Elements
(C) Inline > Elements > Classes > IDs
(D) Classes > IDs > Elements > Inline

45. What is the specificity calculation tuple (a, b, c, d) for the selector 'div#sidebar ul.menu li a:hover'?

(A) (0, 1, 2, 3) - 0 inline, 1 ID, 2 classes/pseudo-classes (.menu, :hover), 3 elements (div, ul, li, a = 4 actually: (0,1,2,4))
(B) (0, 1, 2, 4)
(C) (0, 2, 1, 3)
(D) (1, 1, 1, 1)

46. What is the difference between relative typography units 'em' and 'rem' in CSS?

(A) 'em' is relative to the font-size of its direct parent element; 'rem' (root em) is relative to the font-size of the root <html> element
(B) 'rem' is relative to parent; 'em' is root
(C) 'em' is pixels; 'rem' is points
(D) 'rem' only works on headings

47. What do the viewport units '100vw' and '100vh' represent?

(A) 100% of the browser viewport's width and 100% of the viewport's height respectively
(B) 100 virtual pixels
(C) 100 vector widths
(D) 100 web-safe characters

48. Which CSS combinator matches an element that is a direct immediate child of a specified parent (e.g., 'div > p')?

(A) > (child combinator)
(B) Space (descendant combinator)
(C) + (adjacent sibling)
(D) ~ (general sibling)

49. Which CSS combinator matches an element that immediately follows a specific element as an adjacent sibling (e.g., 'h1 + p')?

(A) + (adjacent sibling combinator)
(B) ~ (general sibling)
(C) > (child combinator)
(D) * (universal)

50. What is the effect of 'position: sticky;' on a header element?

(A) The element behaves as 'relative' until scrolled to a threshold offset, where it toggles to 'fixed' within its container
(B) It remains absolutely fixed at the screen top permanently
(C) It cannot be hidden by opacity
(D) It disables user interaction

51. In Flexbox, what does the shorthand declaration 'flex: 1 1 200px;' represent?

(A) flex-grow: 1, flex-shrink: 1, flex-basis: 200px
(B) flex-direction: 1, flex-wrap: 1, flex-gap: 200px
(C) flex-order: 1, flex-align: 1, flex-width: 200px
(D) flex-columns: 1, flex-rows: 1, flex-margin: 200px

52. In Flexbox, how do you effortlessly center a child element both vertically and horizontally inside a container?

(A) display: flex; justify-content: center; align-items: center;
(B) display: flex; text-align: center; margin: auto;
(C) display: flex; align-content: middle;
(D) display: block; float: center;

53. In CSS Grid, what does 'grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));' accomplish?

(A) Creates a responsive grid where columns automatically wrap and expand to fill space without requiring media queries
(B) Creates exactly 4 fixed columns of 250px
(C) Shrinks all columns to 0 on mobile
(D) Requires 10 media query breakpoints

54. How does 'auto-fit' differ from 'auto-fill' in CSS Grid when there are fewer items than available space?

(A) 'auto-fit' collapses empty tracks and stretches filled tracks to occupy remaining space; 'auto-fill' preserves empty tracks
(B) 'auto-fill' collapses empty tracks
(C) They are 100% identical
(D) 'auto-fit' forces vertical stacking

55. Which CSS property is used to declare CSS Custom Properties (CSS Variables)?

(A) --variable-name: value; (accessed via var(--variable-name))
(B) $variable-name: value;
(C) @variable-name: value;
(D) %variable-name: value;

56. Where are global CSS custom properties standardly declared to ensure inheritance across the entire document?

(A) :root pseudo-class
(B) body selector
(C) * universal selector
(D) html tag

57. Which CSS property provides smooth animated transitions between property state changes?

(A) transition (transition-property, duration, timing-function, delay)
(B) animation
(C) transform
(D) morph

58. Which CSS property enables 2D and 3D visual transformations like rotation, scaling, skewing, and translation?

(A) transform
(B) transition
(C) translate-mode
(D) geometry

59. What rule is used in CSS to define custom multi-step keyframe animation sequences?

(A) @keyframes
(B) @animation
(C) @keyframes-rule
(D) @motion

60. Which CSS property configures keyframe animation duration, iteration count, direction, and fill mode?

(A) animation
(B) transition
(C) transform
(D) play-state

61. What does 'animation-fill-mode: forwards;' ensure after a CSS animation finishes playing?

(A) The target element retains the computed styling values set by the final animation keyframe
(B) The animation restarts immediately in reverse
(C) The element reverts to its original un-animated state
(D) The element becomes hidden

62. What is 'Mobile-First' responsive design strategy in CSS?

(A) Writing base styles for mobile screens first, then using 'min-width' media queries to progressively enhance desktop layouts
(B) Writing desktop styles first and using max-width
(C) Designing native iOS apps before web
(D) Disabling desktop browsers

63. Which CSS pseudo-class matches the first child of its parent container?

(A) :first-child
(B) :first-of-type
(C) :first
(D) :nth-child(0)

64. Which CSS pseudo-class matches every alternate (even) row in a table for zebra-striping?

(A) :nth-child(even) (or :nth-child(2n))
(B) :nth-child(odd)
(C) :alternate-child
(D) :even-row

65. Which CSS pseudo-element is used to insert generated cosmetic content before an element's real content?

(A) ::before
(B) :before-element
(C) ::front
(D) ::prepend

66. What mandatory CSS property must be included inside a '::before' or '::after' pseudo-element rule to render it?

(A) content: "";
(B) display: block;
(C) visibility: visible;
(D) text: "";

67. What causes Margin Collapsing in CSS, and when does it occur?

(A) Vertical margins of adjacent block elements merge into a single margin equal to the maximum of the two margins
(B) Horizontal margins overlap
(C) Padding cancels margin
(D) Borders double margins

68. Which technique establishes a new Block Formatting Context (BFC) to contain floating children and prevent margin collapsing?

(A) 'overflow: hidden;' (or 'display: flow-root;')
(B) 'display: inline;'
(C) 'margin: 0 auto;'
(D) 'position: static;'

69. What is the purpose of the CSS clearfix hack historically used with floated layouts?

(A) To force a parent container to expand and enclose its floated children without collapsing to zero height
(B) To clear browser cookies
(C) To center block elements
(D) To remove margins from images

70. What modern CSS display property replaces the clearfix hack cleanly without unwanted side-effects?

(A) display: flow-root;
(B) display: flex;
(C) display: block;
(D) display: table;

71. Which CSS property allows text to be clipped with an ellipsis (...) when it overflows its container?

(A) text-overflow: ellipsis; (paired with white-space: nowrap and overflow: hidden)
(B) overflow: ellipsis;
(C) word-break: cut;
(D) text-clip: dots;

72. Which CSS property prevents long, unbroken words from overflowing outside their containing boundary?

(A) overflow-wrap: break-word; (or word-break: break-all;)
(B) text-wrap: force;
(C) white-space: break;
(D) word-wrap: clip;

73. Which CSS filter property applies visual blur or drop-shadow effects directly to elements?

(A) filter: blur(5px);
(B) effect: blur(5px);
(C) image-filter: blur;
(D) backdrop: blur;

74. Which modern CSS property blurs the backdrop content behind a translucent glassmorphic card?

(A) backdrop-filter: blur(10px);
(B) background-blur: 10px;
(C) filter: backdrop(10px);
(D) surface-filter: blur;

75. Which CSS feature query rule tests whether a user's browser supports a specific CSS property before applying rules?

(A) @supports (display: grid) { ... }
(B) @if (grid-supported) { ... }
(C) @feature (grid) { ... }
(D) @media (supports-grid) { ... }

76. What is the exact computed specificity value of the selector 'article#main-content div.card:not(.highlight) p > a.link'?

(A) 1 ID, 2 Classes (.card, .highlight via :not), 4 Elements (article, div, p, a) -> (0, 1, 2, 4)
(B) (0, 1, 1, 4)
(C) (0, 2, 2, 3)
(D) (1, 1, 2, 4)

77. In CSS specificity, how does the :is() pseudo-class calculate its specificity compared to :where()?

(A) :is() takes the specificity of its most specific argument; :where() always has a specificity of exactly 0
(B) :where() takes highest specificity; :is() is 0
(C) Both take the average specificity
(D) Both add 10 points

78. What is the Subgrid feature in CSS Grid (grid-template-columns: subgrid;)?

(A) It allows a nested child grid item to inherit and align with the column and row tracks of its parent grid
(B) It compresses grid cells into sub-pixels
(C) It creates a nested table layout
(D) It renders a 3D isometric grid

79. In CSS Flexbox, how is the remaining free space distributed among flex items when their 'flex-grow' values differ?

(A) Proportionally to each item's flex-grow factor multiplied by the available remaining positive free space
(B) Equally regardless of grow factor
(C) By doubling item font-size
(D) By dividing container width by grow sum

80. How does 'flex-shrink' calculate negative free space distribution when items overflow a container?

(A) Proportionally to the product of each item's flex-shrink factor AND its flex-basis width
(B) Purely based on shrink factor alone
(C) Equally across all items
(D) Shrinks only the first item

81. What does the CSS 'contain' property (e.g., 'contain: layout paint;') achieve for browser performance?

(A) Isolates subtrees from the rest of the document, allowing the browser to optimize layout, style, and paint computations independently
(B) Prevents DOM nodes from being deleted
(C) Minimizes JavaScript memory leaks
(D) Compresses CSS stylesheets

82. What is CSS Container Queries (@container) and how does it revolutionize component-driven responsive design?

(A) Allows styling components based on the dimensions of their parent container rather than the overall browser viewport width
(B) Queries backend database containers
(C) Resizes Docker containers via CSS
(D) Replaces media queries entirely

83. What does the CSS 'will-change' property (e.g., 'will-change: transform;') communicate to the browser rendering engine?

(A) Hints that an element will animate, instructing the engine to promote it to its own GPU compositor layer in advance
(B) Forces immediate CPU re-rendering
(C) Pre-loads external fonts
(D) Disables layout recalculations

84. Why can overusing 'will-change' or creating too many GPU compositor layers degrade performance?

(A) Each layer consumes significant GPU VRAM memory; excessive layers cause memory exhaustion and sluggish frame rates
(B) It triggers JavaScript syntax errors
(C) It disables browser caching
(D) It breaks flexbox alignment

85. What is the critical rendering difference between animating 'transform: translate()' versus animating 'top' / 'left' properties?

(A) 'transform' is processed entirely on the GPU compositor thread without triggering CPU Layout or Paint; 'top'/'left' triggers expensive CPU Layout and Repaint on every frame
(B) 'top'/'left' is faster on mobile
(C) 'transform' cannot be smoothed
(D) There is zero performance difference

86. In CSS transitions, what does the 'cubic-bezier(x1, y1, x2, y2)' function define?

(A) A cubic B?zier mathematical curve that defines the acceleration and deceleration rate of an animation over time (timing function)
(B) The 3D coordinates of an element in space
(C) The color gradient transition points
(D) The border curvature radius

87. What does the '@media (prefers-reduced-motion: reduce)' media feature allow developers to respect?

(A) User operating system accessibility settings requesting minimization of non-essential motion, parallax, and flashing animations
(B) Battery saver low-power modes
(C) Slow network connections
(D) Monochrome e-ink displays

88. What does the '@media (prefers-color-scheme: dark)' media query enable?

(A) Applying dark-themed stylesheets automatically when a user configures dark mode at the OS level
(B) Testing monitor color calibration
(C) Detecting ambient room light
(D) Inverting RGB colors via hardware

89. In CSS Grid, what happens when an item has 'grid-column: 1 / -1;'?

(A) The item spans across all column tracks from the first explicit line to the very last line of the grid
(B) It causes an index out of bounds error
(C) It centers the item in the first column
(D) It hides the item

90. What is the CSS 'clamp(min, preferred, max)' mathematical function commonly used for in modern responsive web design?

(A) Fluid typography and spacing: dynamically scales between min and max boundaries based on viewport width without media queries
(B) Clamping negative numbers to zero
(C) Restricting color saturation
(D) Binding image aspect ratios

91. In the CSS rule 'font-size: clamp(1rem, 2.5vw, 2rem);', what is the rendered font size on a 1000px wide viewport (assuming 1rem = 16px)?

(A) 25px (2.5vw = 25px; fits safely between min 16px and max 32px)
(B) 16px
(C) 32px
(D) 20px

92. What is CSS Specificity War, and how does the CSS Cascade Layers (@layer) feature resolve it cleanly?

(A) Cascade Layers allow developers to explicitly control layer precedence order regardless of selector specificity within layers
(B) Increases specificity by chaining IDs
(C) Deprecates the !important flag
(D) Disables external stylesheets

93. What does the CSS 'pointer-events: none;' property do to an element?

(A) Makes the element completely transparent to mouse, touch, and click events, passing them through to elements underneath
(B) Hides the mouse cursor
(C) Disables CSS animations
(D) Prevents right-clicking on images

94. What is the purpose of the CSS 'touch-action: manipulation;' property on mobile web applications?

(A) Disables double-tap to zoom on interactive buttons, eliminating the legacy 300ms mobile click delay
(B) Enables multi-touch pinch zooming
(C) Rotates the screen on swipe
(D) Simulates haptic vibration feedback

95. Which CSS property maintains an element's dimensional proportions automatically (e.g., 16/9 for responsive video)?

(A) aspect-ratio: 16 / 9;
(B) dimensions: 16-9;
(C) ratio: 1.77;
(D) scale-ratio: video;

96. What is the 'object-fit: cover;' property used for when styling <img> or <video> elements?

(A) Resizes the content to cover the entire container box while preserving aspect ratio, clipping overflowing portions cleanly
(B) Stretches image horizontally to distort it
(C) Shrinks image to fit inside without cropping
(D) Converts raster image into SVG

97. How does 'object-fit: contain;' differ from 'object-fit: cover;'?

(A) 'contain' scales the media to fit entirely inside the container box without cropping, potentially leaving letterbox pillar bars
(B) 'contain' always crops images
(C) 'contain' stretches images unevenly
(D) 'contain' disables responsive scaling

98. What does the CSS pseudo-class ':focus-visible' achieve for keyboard vs mouse user accessibility?

(A) Applies focus outline rings ONLY when navigation occurs via keyboard (Tab key) or assistive tools, suppressing rings on mouse clicks
(B) Hides focus indicators completely
(C) Changes focus color to neon green
(D) Locks keyboard focus inside an iframe

99. What is the difference between CSS pseudo-class ':has()' (the parent selector) and traditional selectors?

(A) ':has()' matches a parent element based on whether it contains descendants matching a selector (e.g., 'figure:has(figcaption)')
(B) It is an alias for :not()
(C) It only selects child elements
(D) It selects sibling text nodes

100. What is the role of CSS Preprocessors (such as Sass / SCSS) in professional web engineering workflows?

(A) Provides variables, nesting, mixins, functions, and modular partial imports that compile into standard browser-compatible CSS
(B) Executes CSS directly on backend servers
(C) Replaces JavaScript for DOM operations
(D) Minifies HTML documents

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the CSS Box Model in comprehensive technical detail, contrasting standard box-sizing: content-box with border-box.

Q2. Analyze CSS Specificity rules, specificity calculation tuples (a, b, c, d), the Cascade, and inheritance mechanics.

Q3. Compare CSS positioning schemes: static, relative, absolute, fixed, and sticky with architectural use cases.

Q4. Explain the Flexbox 1-dimensional layout model in complete detail: axes, container properties, and item alignment.

Q5. Explain the CSS Grid 2-dimensional layout architecture: grid tracks, grid lines, grid areas, fr units, and minmax().

Q6. Compare and contrast CSS Flexbox with CSS Grid, formulating explicit engineering criteria for when to deploy each.

Q7. Explain Responsive Web Design (RWD) methodologies, Mobile-First vs Desktop-First philosophies, and Media Queries.

Q8. Explain CSS Transitions and Animations, detailing transition-timing-function, @keyframes rules, and GPU acceleration.

Q9. Explain CSS Custom Properties (CSS Variables), their declaration scope, cascade inheritance, and dynamic JavaScript manipulation.

Q10. Describe CSS Container Queries (@container) and contrast them with traditional viewport-based Media Queries (@media).

Q11. Explain the browser Critical Rendering Path (CRP) and how CSS styles impact DOM/CSSOM construction, Layout, and Paint.

Q12. Detail CSS Typography units: px, pt, em, rem, vw, vh, and explain fluid typography using clamp().

Q13. Explain CSS Combinators (Descendant, Child, Adjacent Sibling, General Sibling) with code snippets and rendering rules.

Q14. Describe CSS Pseudo-classes and Pseudo-elements, detailing the differences between structural, state, and cosmetic selectors.

Q15. Explain Margin Collapsing mechanics in CSS and outline all four conditions that prevent margins from collapsing.

Q16. Explain CSS Cascade Layers (@layer) and illustrate how they solve large-scale enterprise stylesheet specificity conflicts.

Q17. Describe CSS Subgrid architecture and demonstrate how it aligns nested child items with an outer parent grid.

Q18. Detail CSS filter and backdrop-filter properties, explaining glassmorphism, blur, saturation, and contrast effects.

Q19. Explain CSS Dark Mode implementation strategies using CSS Custom Properties and prefers-color-scheme media query.

Q20. Discuss CSS Accessibility best practices: :focus-visible outlines, color contrast ratios (WCAG AA/AAA), and prefers-reduced-motion.

Q21. Write a complete CSS Grid layout rule set for a standard responsive blog template: Header, Sidebar, Main Content, and Footer.

Q22. Write a comprehensive Flexbox navigation bar stylesheet featuring logo on the left, centered menu links, and call-to-action buttons on the right.

Q23. Explain CSS Transform 3D coordinate space and write a complete CSS stylesheet for a 3D perspective flip-card animation.

Q24. Detail the CSS pseudo-class :has() and provide practical architectural examples of parent styling based on child states.

Q25. Explain the implementation of CSS Scroll Snap points for creating carousel and full-page touch sliders.

Q26. Discuss CSS Print Stylesheets (@media print), explaining page breaks, color optimization, and link URL expansions.

Q27. Explain the architecture of CSS Preprocessors (SCSS/Sass), detailing Variables, Nesting, Mixins, Partials, and Inheritance.

Q28. Describe CSS Multi-Column Layout module properties: column-count, column-gap, column-rule, and column-span.

Q29. Explain CSS Shape Outside (shape-outside) and demonstrate how to wrap text fluidly around circular or polygon images.

Q30. Analyze common CSS layout bugs: Horizontal Scrolling Overflow, Flex item shrinking quirks, and z-index Stacking Context traps.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

The !important Override: Appending !important overrides normal specificity cascades and should be used sparingly to avoid unmaintainable specificity wars.

>>Solution Question Bank The CSS box model is a fundamental concept that describes how elements are rendered on a web page. It consists of four components: content, padding, border, and margin. Each component has its own properties that can be set using CSS. The box model is important because it affects the layout and spacing of elements on a page. Understanding the box model is essential for creating responsive and visually appealing web designs. It allows developers to control the size and position of elements, as well as their relationship with other elements on the page. By manipulating the box model properties, developers can create complex layouts and achieve precise alignment.

6. gap: Sets gutter spacing between flex children.

Flex Item Properties:
- flex-grow: Proportion of available remaining space the item should absorb.
- flex-shrink: Ability of the item to shrink when space is constrained.
- flex-basis: Default starting size of the item before remaining space is distributed.
- align-self: Overrides container align-items for a specific individual child.

>>Solution Question Bank: Mastering CSS Grid 2. CSS Flexbox and CSS Grid for Creating Responsive and Complex Layouts

components where item content sizes vary.
- Grid is Layout-First: The container establishes rigid structural tracks into which child items are placed.

Engineering Decision Criteria:
- Use Flexbox for: Navigation bars, form button groups, media object cards, centering a single modal, and distributing linear UI components.
- Use CSS Grid for: Overall webpage scaffolding (header, sidebar, main, footer), image galleries, dashboard card layouts, and complex overlapping multi-column layouts.

>>Solution Question Explain Responsive Web Design (RWD) methodologies. Mobile-First vs Desktop-First approaches, and CSS Media Queries.

```css
@keyframes pulseGlow {
0% { transform: scale(1); opacity: 0.8; }
50% { transform: scale(1.08); opacity: 1; }
100% { transform: scale(1); opacity: 0.8; }
}
.pulse-button {
animation: pulseGlow 2s infinite ease-in-out;
}
```

GPU Acceleration & Performance:
Animating geometry properties (width, height, top, left) forces CPU Layout and Paint cycles on every frame (causing frame drops). Animating transform (translate, scale, rotate) and opacity runs directly on the GPU compositor thread, guaranteeing butter-smooth 60/120 FPS animations.

>> Solution to Question 10: Explain CSS Custom Properties (CSS Variables), their declaration scope, cascade inheritance, and dynamic JavaScript manipulation.

[Problem Statement]: Describe CSS Container Queries (@container) and contrast them with traditional viewport-based Media Queries (@media)
[Detailed Step-by-Step Answer]:
While Media Queries inspect the global browser viewport, modern modular component architecture demands styling based on the component's immediate surrounding container:

The Container Queries Revolution (@container):
Consider a responsive Product Card component. When placed in the wide main content column, it should render horizontally with an image on the left. When the exact same component is placed inside a narrow sidebar, it must render vertically stacked.

Implementation Steps:
1. Declare Container Context on Parent:
```css
.card-wrapper {
container-type: inline-size;
container-name: cardContainer;
}
```
2. Query Container Dimensions on Child Component:
```css
@container cardContainer (min-width: 450px) {
.card {
display: flex;
flex-direction: row;
}
}
```
This completely decouples component responsiveness from viewport screen dimensions, enabling truly modular, reusable web UI widgets.

>> **Solution to Question 11:** [Problem Statement]: Explain the browser Critical Rendering Path (CRP) and how CSS styles impact DOM/CSSOM construction, Layout, and Paint.

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>>Solution Question 12: CSS Combinators (Descendant, Child, Adjacent Sibling, General Sibling) with code snippets and rendering rules.

```
/* 1. Declare layer precedence order (lowest to highest) */
@layer reset, base, framework, components, utilities;

@layer framework {
/* High specificity ID selector inside lower layer */
#nav.primary .item { color: black; }
}

@layer components {
/* Low specificity class selector inside higher layer */
.item { color: blue; }
}
```

Resolution: Even though `#nav.primary .item` has a specificity of (0,1,1,1) in the framework layer, `.item` in the components layer WINS because declarations in later layers always defeat declarations in earlier layers regardless of internal specificity!

>>Solution to Question 15: CSS Subgrid allows items to align with a parent grid's tracks. **Question 16:** How does CSS Scroll Snap work?

Modifies visual rendering of an element and all of its children:
`filter: blur(4px) brightness(1.2) contrast(90%) drop-shadow(0 4px 8px rgba(0,0,0,0.3));`

2. The 'backdrop-filter' Property (Glassmorphism):
Applies visual filters to the graphical content BEHIND a translucent element without blurring the element's own text or icons:

```css
.glass-modal {
background: rgba(255, 255, 255, 0.25);
backdrop-filter: blur(12px) saturate(180%);
border: 1px solid rgba(255, 255, 255, 0.3);
border-radius: 16px;
}
```

This creates modern, frosted-glass UI aesthetics popular in modern desktop and mobile operating system interfaces.

>>Solution to Question 19: Explain CSS Dark Mode implementation strategies using CSS Custom Properties and prefers-color-scheme media query.

```
  background-color: var(--bg-primary);
  color: var(--text-primary);
  }
  ```
```

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>>Solution Question Bank Access Provided by best universities for assistance and learning.
```

```
"sidebar"
"footer";
gap: 16px;
}

@media (min-width: 768px) {
.site-layout {
grid-template-columns: 280px 1fr;
grid-template-areas:
"header header"
"sidebar main"
"footer footer";
}
}

header { grid-area: header; }
aside { grid-area: sidebar; }
main { grid-area: main; }
footer { grid-area: footer; }
```

>>Solution to Question 22: a comprehensive Flexbox navigation bar stylesheet featuring logo on the left, centered menu links, and call-to-action buttons on t

```
  .nav-links a:hover { color: #0066cc; }

  .nav-actions { display: flex; gap: 12px; }
```

>>Solution to Question 28: The CSS transform-style property is used to create a 3D space for its children. Example: .parent { transform-style: preserve-3d; } Explanation: Without this property, child elements would be flattened onto the parent's plane, losing the 3D effect.

```
.form-group:has(input:invalid:focus) {
border-color: #ff3333;
background-color: #fff5f5;
}
```

2. Styling Card Headers When Card Contains Images:

```css
.card:has(img) .card-header {
position: absolute;
background: rgba(0,0,0,0.5);
color: white;
}
```

3. Navigating Sibling State Without JavaScript:

`body:has(#nav-toggle:checked) { overflow: hidden; }` (Locks scrolling when mobile menu drawer is open).

```
}

/* 2. Optimize text contrast and save printer ink */
body {
color: #000000;
background: #ffffff;
font-size: 12pt;
}

/* 3. Expand hyperlinks so readers can see destinations */
a[href]::after {
content: " (" attr(href) ")";
font-size: 9pt;
}

/* 4. Manage page break breaks */
h1, h2 { break-after: avoid; }
table, figure { break-inside: avoid; }
}
```

>>Solution Question 25: Describe the CSS Media Rule. CSS Preprocessors (SCSS/Sass) detail Variables, Nesting, Mixins, Modules, and Inheritance.

(shorthand `columns: 250px 3;`).
3. column-gap: Sets the gutter clearance between columns.
4. column-rule: Draws a vertical dividing rule line between columns (e.g., `column-rule: 1px solid #ccc;`).
5. column-span: Allows elements (like major headings) to break out of multi-column flow and span across all columns (`column-span: all;`).

>>Solution Question Bank CSS Shapes CSS Layout (Inline-block, Floats, Flexbox, Grid) Overflow Text truncation & wrapping Z-index & Stacking Context traps.